

COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

*Machine-computed from 16 years of daily price data through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. **These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. This cycle's active qualitative score set is 2026-09-26_draft2, a FULL WEEKLY refresh — Fed & Macro was re-rated DOWN on a genuinely new primary-sourced cluster (Fed Governor Barr's Sept 23 hawkish speech plus the same-day S&P Global flash PMI) that the intervening daily touches had missed; CLARITY Act, Iran War Risk and Tokenization were each independently re-researched and HELD FLAT — LEFT AS DRAFT, AWAITING ALEX'S MANUAL REVIEW (full weekly refreshes are categorically excluded from delegated auto-stamp authority). See those pillar reports for the full research trail. This quant pillar's own output is unaffected by qualitative status and enters the master blend the same way regardless.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through September 25, 2026 (BTC \$83,381.75) · Pipeline data refreshed via GitHub Actions; digest read via run.py numbers & verified September 26, 2026 (qualitative set: FULL WEEKLY refresh, 2026-09-26_draft2, draft, awaiting Alex's stamp) · Not Financial Advice

The Scores — September 25 Model Output

Computed September 25, 2026 (previous computed run September 23, 2026) — same standard as every qualitative pillar. Weekly, monthly, and quarterly bars all remain PARTIAL this cycle; both as-is and completed-bar readings are quoted together throughout.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	58	0.75	8	+0.96	Mild Bull — the weekly close above the 200-week MA remains confirmed; the newly-forming week reads +53.9 as-is, +46.5 on completed bars; 1mo analog reads 1-of-3 positive, median −1.8%
3 months	61	0.70	8	+1.23	Mild Bull — the 3mo analog reads unanimous 3-of-3 positive, median +23.0%
6 months	61	0.55	8	+0.97	Mild Bull — unanimous 3-of-3 positive, median +78.6%
1 year	68	0.60	8	+1.73	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3), median +96.3%
3 years	69	0.50	8	+1.52	Bull — value + cycle (36-month analog 3-of-3 positive, median +1421.9%)

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P − 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

- P** — the odds the tape is *good for Bitcoin* by then. Near-term still confirmed on a completed-bar basis, and the newly-forming week reads \$83,381.75 (weekly composite as-is +53.9/100, Moderate bullish, agreement 6-of-7; completed-bar weekly composite holds at +46.5); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, median +96.3%, and price sits ~41% below its power-law fair value).
- Quality** — how trustworthy the evidence is. Highest near-term (weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).
- Impact** — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

- The crowd:** this cycle is a light daily touch — Iran War Risk was re-rated DOWN, reversing the Sept 23 bump, on WSJ reporting (widely syndicated) that Trump rejected Iran's Hormuz proposal; CLARITY Act, Tokenization RWA and Fed & Macro were checked fresh and held flat — and the tape itself pulled back since this pillar's last refresh. (Price sits at \$83,381.75, down from \$85,488.24 on Sept 23 — a −2.5% pullback, and +42.5% off the June 30 low; the tape doesn't know why price moved, only that it did.)
- The model:** the short-term tape eased with the price move — the weekly composite (as-is) reads Moderate bullish (+53.9, agreement 6-of-7), the monthly composite reads Neutral (+15.1, agreement 3-of-7), and the quarterly composite reads Slight bullish (+30.8, agreement 5-of-7). All three bars (weekly, monthly, quarterly) remain unclosed this cycle, so their completed-bar composites are unchanged: weekly +46.5, monthly +1.8, quarterly +7.7. Repair Signal #1 (a weekly close above the 200-week MA) stays confirmed, though price eased against the 50-week MA

(\$78,156, +6.7%, down from +9.3%). The 1-month window holds Mild Bull (P 58) with its cycle analog at 1-of-3 positive (median -1.8%); 3 months holds Mild Bull (P 61) as its analog strengthened further (3-of-3 positive, median +23.0%, up from +17.7%). The 6-month window reads Mild Bull (P 61, down one point) with the analog still unanimous (3-of-3 positive, median +78.6%). The long read sits at 1yr (68, up one point) and 3yr (69, up one point): every prior cycle at this exact month was higher 12 months later (median +96.3%), price is ~41% below fair value (\$141,180), and the froth that precedes real crashes remains absent.

The edge: the model reads three separate clocks that don't take a position on the news cycle — tape (confirmed at the weekly level, easing with today's pullback), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 29.2, past the average-bottom date), and valuation (~41% below the power-law fair line, with price sitting above the -1σ "cheap" band at \$70,521). All three clocks point the same direction on the medium-to-long horizon — the near-term clock is the one doing the day-to-day flexing, exactly as its lower Quality input says it should.

The Facts — The Tape

*Computed September 25, 2026 from daily data (BTC \$83,381.75, daily close as of September 25). Weekly, monthly, and quarterly bars are all still forming — scores use them as-is. Quantified: on completed bars only, weekly is **+46.5** (bar closing 2026-09-27), monthly +1.8 and quarterly +7.7 (both bars closing 2026-09-30) — all unchanged from Sept 23. Both readings are quoted deliberately — the as-is figure includes a bar that has not finished.*

- **Weekly composite +53.9/100 (Moderate bullish), agreement 6-of-7 indicators** — eased with price; on completed bars only it reads +46.5, and both must be quoted together because the current bar is still forming. Price is above the 20-week (\$70,267, +18.7%) and well above the **200-week average (\$65,831, +26.7%)**; price sits above the 50-week (\$78,156, +6.7%). RSI 60.7; MACD line +784 with histogram positive (+2,723) — momentum repair continuing; %B 0.93.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) remains CONFIRMED.** Price (\$83,381.75) sits +26.7% above the 200-week average (\$65,831). The weekly bar remains unclosed, still forming, closing 2026-09-27; the completed-bar weekly composite holds at **+46.5**.
- **Monthly composite +15.1/100 (Neutral), agreement 3-of-7** — eased with price. Below the 20-month average (\$87,270, -4.5%), above the 50-month (\$62,253, +33.9%); RSI 53.2; monthly MACD line +705 with histogram -3,419 — still negative; %B 0.44. Completed-bar monthly composite (August's closed bar) holds at +1.8, unchanged.
- **Quarterly composite +30.8/100 (Slight bullish), agreement 5-of-7** — eased with price. Comfortably above the 20-quarter average (\$58,391, +42.8%) and the 50-quarter (\$28,833, +189.2%); RSI 58.4, MACD line +16,614 with histogram -123 — turned negative this cycle. Completed-bar quarterly composite is +7.7, unchanged (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price (\$83,381.75) sits just below overhead resistance at \$89,164 (+6.9%, 1x touch) — the nearest test on the board — then \$98,090 (+17.6%, 2x) and \$108,330 (+29.9%, 3x). Support sits at \$81,685 (-2.0%, 2x touch), then \$74,434 (-10.7%, 1x) and \$59,930 (-28.1%, 1x) further below. The falling weekly support trendline sits at \$40,880; the falling resistance trendline sits at \$73,991, now well below current price.
- ★ **June 30 remains the cycle low: \$58,526 close** — the flush tagged the Fib golden pocket (\$54,257–\$57,802 = 61.8–65% retrace of the \$15,479 → \$126,272 run) and has since built a **+42.5% bounce to \$83,381.75**. **\$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.



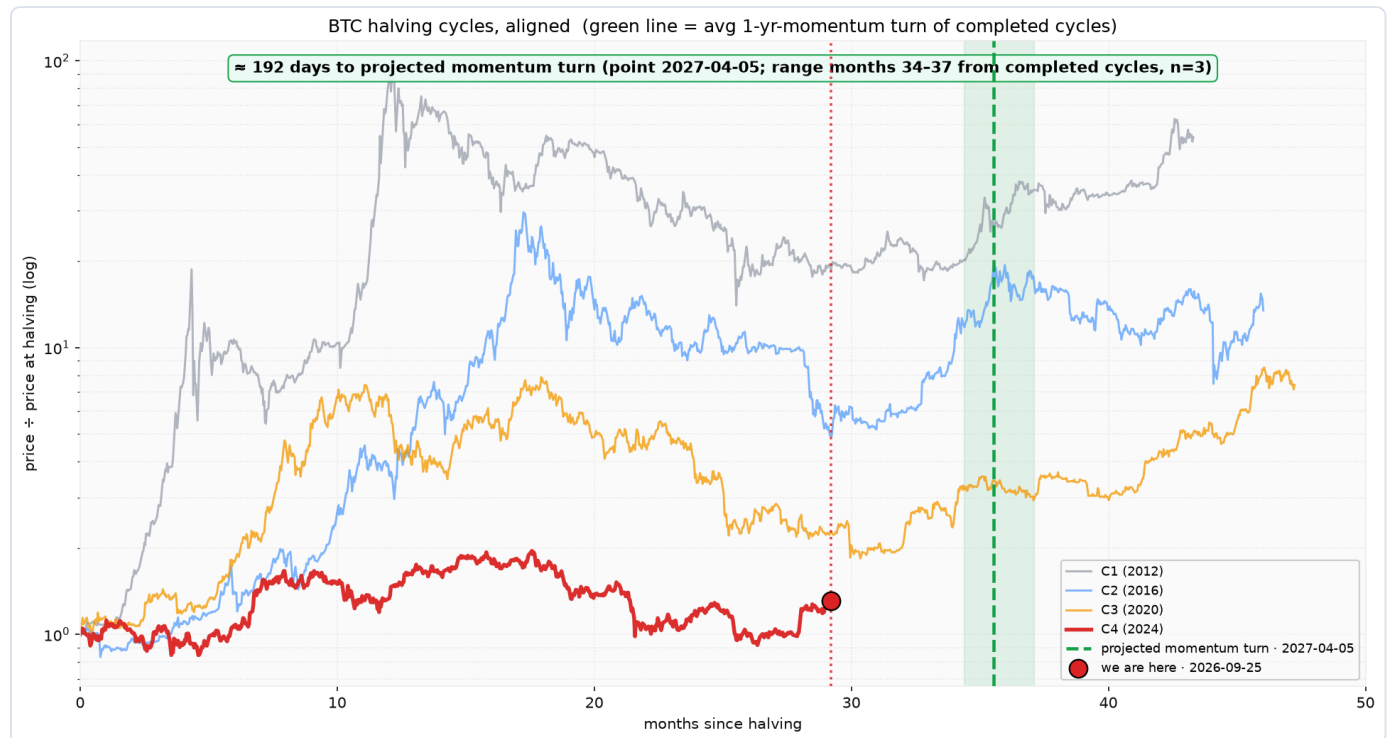
Weekly tape, scored +53.9/100 as-is (Moderate bullish, +46.5 on completed bars only) with agreement 6 of 7 indicators: the weekly close above the 200-week MA remains confirmed, MACD histogram remains positive, price above the 50-week MA. Machine-annotated from the same data that feeds the scores.

The Facts — The Cycle Clock & The Value Floor

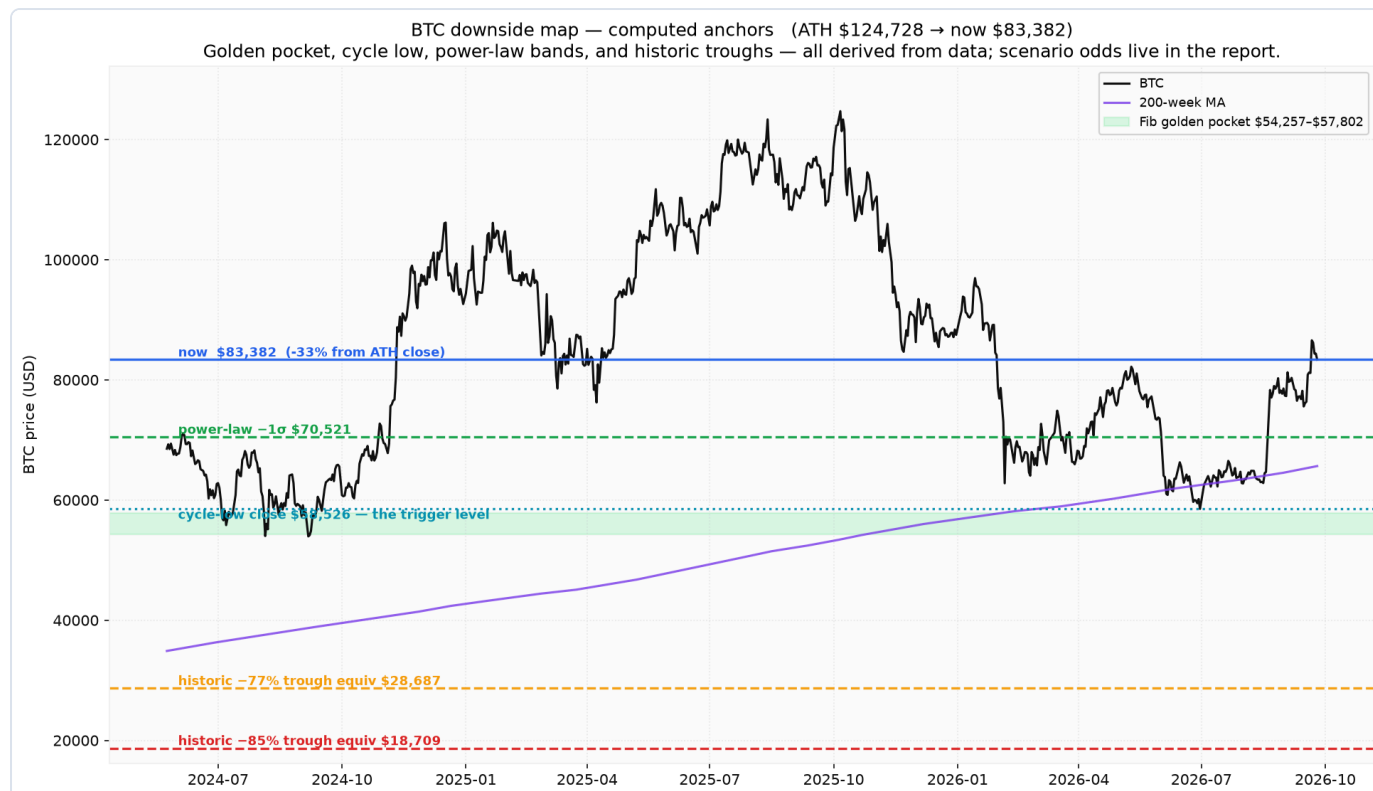
The two slower clocks. Same data, longer lenses.

- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — \$124,728) and printed its low-so-far at month 26 (June 30, \$58,526, –53% from the top). **We are at month 29.2 — past the average-bottom date.**
- ★ **The cycle analogs are the spine of the long-window scores.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **1-of-3** (median –1.8%) · next 3 months positive **3-of-3** (median +23.0%) · next 6 months positive **3-of-3** (median +78.6%) · next 12 months positive **3-of-3** (median +96.3%) · next 36 months positive **3-of-3** (median +1421.9%). *Fragility disclosure: the medians and agreement counts move as the cycle-month clock advances; treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~29 that window reaches month ~65 — it contains the NEXT halving's rally.*
- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5) → still projects to ~April 2027 for this cycle.**
- **The power-law floor:** a log-log fit over the full price series puts fair value at **\$141,180** — price is **–41%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value materially (\$130,853 / \$135,767 / \$118,950 / \$145,806) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price (\$83,381.75) sits above the **–1σ band (\$70,521)** — out of the "cheap" zone (the –2σ band sits at \$35,226, far below). Every halving-era cycle trough was –77% to –85% from ATH (\$18.7K–\$28.7K equivalent today) — the pre-halving 2011 bear went ~–93% (~\$8.7K equivalent), excluded from the range but in the data; this cycle's low-so-far was –53% from the ATH.

- **No froth in the system:** the altcoin-froth index reads $z = -0.87$ — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to -39% average 12-month drawdowns vs -23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23, unchanged)* — used as a tilt, not a score input.
- ★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 29.2, past the historical bottoming window.



The downside map, updated after the June 30 flush: the projected \$54K–\$61K bottom zone was hit (\$58,526 close / \$57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to \$54–\$58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From \$83,381.75: the 20-week average (~\$70.3K) is well behind us — price sits **+18.7%** above it; the 50-week average (\$78,156) is further behind, price sitting **+6.7%** above it. A retest of the \$124,728 ATH is **+49.6%**; the power-law fair line is **+69.3%**. The median 12-month analog from this cycle-month is now **+96.3%**; the median 36-month analog is +1421.9% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the near-term signal remains confirmed on a completed-bar basis and holds Mild Bull (1mo P 58) in a cycle where the only qualitative move was Iran War Risk's re-rate back down (reversing its Sept 23 bump) — this stretch's price pullback is the tape's own, not that catalyst's; the near quarter holds Mild Bull (3mo P 61, analog strengthening further), and the 6-month-and-beyond case remains unanimous (3-of-3) and firm (6mo P 61, 1yr P 68, 3yr P 69).

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A -53% drawdown at month 26 vs -77/-85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is \$18.7K–\$28.7K, far below every level in this report.
- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.

- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
\$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket \$54.3–57.8K is the target; re-rate 1/3-month down
~\$65,831	Weekly close above the 200-week MA	CONFIRMED — price sits +26.7% above this level; the weekly bar remains unclosed, still forming, closing Sept 27; the completed-bar weekly composite holds at +46.5
~\$78,156	Weekly close above the 50-week MA	CONFIRMED — price sits +6.7% above it, trend repair intact despite today's pullback
~Aug 31, 2026	Historical average-bottom month (28.4) has now passed (we're at month 29.2)	Price sits well off the June 30 low (+42.5%); a higher low + improving monthly momentum after this date = basing confirmed — August's completed monthly bar (+1.8) is an early such sign
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
\$50,000	Weekly close below	–60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **\$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone (\$18.7K–\$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: DRAFT, AWAITING ALEX'S STAMP (2026-09-26_draft2, FULL WEEKLY REFRESH) · THIS PILLAR: COMPUTED SEPTEMBER 25, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 58 / 61 / 61 / 68 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8 · This pillar's signals (unchanged): 1mo +0.96 · 3mo +1.23 · 6mo +0.97 · 1yr +1.73 · 3yr +1.52 · Master (equal-weight), as of this cycle's blend: 1mo -2.10 · 3mo -1.63 · 6mo -0.59 · 1yr +1.13 · 3yr +2.81 · THE FINAL NUMBER (this blend): -0.07 equal-weight (Neutral) / +0.47 horizon-weighted (Neutral) · Leave-one-out without this pillar: -0.41 (equal-weight) — see the other four pillar reports for their own independent Sept 26 findings; Fed & Macro drove this cycle's move

Sept 25 data refresh, this version: this is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. Weekly, monthly, and quarterly bars all remain unclosed this cycle (completed-bar composites unchanged: weekly +46.5, monthly +1.8, quarterly +7.7), and the as-is weekly composite eased to +53.9 (6-of-7) with today's pullback. The 1-month window holds Mild Bull (P 58) with its cycle analog at 1-of-3 positive (median -1.8%) as the cycle-month clock advanced to 29.2. The 6-month window ticked down one point to Mild Bull (P 61) with its analog still unanimous (3-of-3 positive, median +78.6%). The 3-month window holds Mild Bull (P 61), its analog strengthening further (3-of-3 positive, median +23.0%, up from +17.7%). The long read ticked up one point each to 1yr (68) and 3yr (69) — the direction, not the level, is the signal. This is a purely mechanical result of the updated price/indicator/cycle-analog/power-law picture; BTC pulled back from \$85,488.24 (Sept 23) to \$83,381.75 (Sept 25), a real market move, not a qualitative research finding, and this pillar is **never hand-edited** — config/scores.json never carries this pillar's numbers, which are computed by the pipeline straight into data/pillar_scores.json.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step, and no auto-stamp decision to make (it carries no draft/committed status at all). Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: this cycle's active qualitative score set is 2026-09-26_draft2, a FULL WEEKLY refresh — Fed & Macro was re-rated DOWN on a genuinely new primary-sourced cluster (Fed Governor Barr's Sept 23 hawkish speech plus the same-day S&P Global flash PMI) that the intervening daily touches had missed; CLARITY Act, Iran War Risk and Tokenization were each independently re-researched and HELD FLAT — LEFT AS DRAFT, AWAITING ALEX'S MANUAL REVIEW (full weekly refreshes are categorically excluded from delegated auto-stamp authority). This pillar's own computed output is unaffected by that process and enters the master blend the same way it does every cycle.

Prior version — Sept 23 data refresh (P 58/61/62/67/68): weekly, monthly, and quarterly bars all remained unclosed that cycle (completed-bar composites: weekly +46.5, monthly +1.8, quarterly +7.7), and the as-is weekly composite eased slightly to +59.0 (6-of-7) with that cycle's pullback. The 1-month window held Mild Bull (P 58) with its cycle analog at 1-of-3 positive (median -5.9%) as the cycle-month clock advanced to 29.14. The 6-month window read Mild Bull (P 62) with its analog unanimous (3-of-3 positive, median +59.1%). The 3-month window firmed to Mild Bull (P 61), its analog strengthening (3-of-3 positive, median +17.7%, up from +12.2%). The long read sat at 1yr (67) and 3yr (68). That cycle's active qualitative score set was 2026-09-23_draft, a LIGHT DAILY touch — Iran War Risk re-rated up on two Sept 22 diplomatic developments; CLARITY Act, Tokenization RWA and Fed & Macro carried forward unchanged — COMMITTED, AUTO-STAMPED SEPTEMBER 23, 2026.

What Changed Since Last Check

Pipeline data refreshed September 25, 2026 via GitHub Actions (previous computed run September 23, 2026).

- **Today's refresh in one line:** this is a light daily touch syncing this pillar's price/indicator snapshot to current, fully automated — a computed refresh, not a research judgment call. BTC moved from \$85,488.24 (Sept 23) to **\$83,381.75** (Sept 25), a -2.5% pullback (a real market move, not a qualitative finding), and the model P moved

- from 58/61/62/67/68 to **58/61/61/68/69** — 6mo ticked down one point, 1yr and 3yr each ticked up one point; small, mechanical moves from the updated technical picture, not a judgment call.
- **No data revision this cycle:** this cycle read the current GitHub Actions relay data already committed to the repo — no conflicting prior-close values to disclose.
 - **Price: \$85,488.24 (Sept 23) → \$83,381.75 (Sept 25).** A -2.5% pullback — price eased against the 50-week MA; this pillar scores the tape, not the day's qualitative news, and this move is the tape's own, unrelated to Iran War Risk's own re-rate this cycle (see that report).
 - ★ **Repair signal #1 remains CONFIRMED:** price sits +26.7% above the 200-week average (\$65,831), despite today's pullback.
 - **Weekly, monthly, and quarterly bars all remain unclosed this cycle.** Completed-bar composites: weekly **+46.5**, monthly **+1.8**, quarterly **+7.7** — all unchanged from Sept 23. The now-forming (as-is) composites read: weekly **+53.9** (agreement 6/7, down from +59.0, bar closes Sept 27), monthly **+15.1** (agreement 3/7, down from +19.2, still Neutral), quarterly **+30.8** (agreement 5/7, down from +32.6). Both readings must be quoted together because the current bars are PARTIAL.
 - **Model P vs Sept 23: 58/61/62/67/68 → 58/61/61/68/69.** The cycle-month clock advanced to **29.2**; the 1-month analog reads 1-of-3 positive (median -1.8%), the 3-month analog strengthened further (3-of-3 positive, median +23.0%, up from +17.7%), and the 6-month analog remains unanimous (3-of-3, median +78.6%) though the window's P ticked down one point on the updated mapping. Honesty note, unchanged: analogs rest on **n=3 completed cycles** and single windows flip when alignment shifts by days — the direction is the signal, the levels are soft.
 - **Power-law fair value: \$140,999 (Sept 23) → \$141,180 (Sept 25)** (price now -41% vs. fair, from -39%, on the price pullback) — normal drift as new daily bars extend the fit. Fit-window sensitivity this cycle spans \$118,950 (2013 start) to \$145,806 (2015 start): quote it as a range, never a target.
 - **Froth unchanged at z = -0.87**, still dated 2026-05-23 — Coin Metrics community data lags ~6 weeks, so this stays a robustness tilt, never a score input.
 - **Data refreshed:** daily price data through September 25, 2026 via the GitHub Actions relay; indicators/levels/scores/cycle/mapping read via `run.py numbers`.
 - **Context:** this cycle's active qualitative score set is **2026-09-26_draft2**, a FULL WEEKLY refresh, LEFT AS DRAFT awaiting Alex's manual review, in which Fed & Macro was re-rated DOWN on a genuinely new primary-sourced cluster (Fed Governor Barr's Sept 23 hawkish speech plus the same-day S&P Global flash PMI) that the intervening daily touches had missed, while CLARITY Act, Iran War Risk, and Tokenization RWA were each independently re-researched and held flat. This quant pillar is computed fresh every run regardless of qualitative status, and is never hand-edited: `config/scores.json` never carries this pillar's numbers, which are written by the pipeline straight to `data/pillar_scores.json`.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via [fetch_btc_history.py](#) · Coin Metrics community reference price (2010–2011 close-only) via [build_master_history.py](#) — 5,838 days total, each row tagged by source.
- **Indicator stack:** [indicators.py](#) — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. [levels.py](#) — swing-cluster S/R + least-squares trendlines.
- **Scoring:** [score.py](#) — each indicator → conviction in [-100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** [cycle_analysis.py](#) — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · [downside.py](#) / [downside_map.py](#) — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · [test_altcoin_froth.py](#) — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** [pillar_mapping.py](#) — window scores from tape composites + cycle analogs + power-law value; $P = 50 + 25 \times (S/100)$; Quality rubric per window; writes `data/pillar_scores.json` for the master blend.

- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** weekly/monthly/quarterly bars are all partial as of Sept 25 (completed-bar composites: weekly +46.5, bar closing Sept 27 / monthly +1.8, August's bar having closed / quarterly +7.7, current bar still forming) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle claims, and analog windows can cross into the next cycle · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.